

A Project Report
On
SamvaadX: Real-Time Chat Application Using
Node.js and Socket.

BACHELOR OF TECHNOLOGY



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By
ANKIT PATEL
23131010430

Under the guidance of
DR. SUVEG MOUGDIL

SCHOOL OF COMPUTING SCIENCE AND ENGINEERING DEPARTMENT OF
COMPUTER SCIENCE AND ENGINEERING

GALGOTIAS UNIVERSITY, GREATER NOIDA

INDIA

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**SCHOOL OF COMPUTING SCIENCE AND
ENGINEERING**
GALGOTIAS UNIVERSITY, GREATER NOIDA

CANDIDATE'S DECLARATION

I/We hereby certify that the work which is being presented in the project, entitled “**SamvaadX**” in partial fulfillment of the requirements for the award of the B. Tech. in Computer Science and Engineering submitted in the School of Computing Science and Engineering of Galgotias University, Greater Noida, is an original work carried out during the period of February 2026 to June 2026, under the supervision of **Prof. SUVEG MOUDGIL** Department of Computer Science and Engineering, of School of Computing Science and Engineering , Galgotias University, Greater Noida.

The matter presented in the thesis/project/dissertation has not been submitted by me/us for the award of any other degree of this or any other places.

ANKIT PATEL

23131010430

CERTIFICATE

This is to certify that Project Report entitled “**ANKIT PATEL**” which is submitted by **DR. SUVEG MOUGDIL** in partial fulfillment of the requirement for the award of degree B. Tech. in Computer Science and Engineering of Department of Computer Science and Engineering, School of Computing Science and Engineering, Galgotias University, Greater Noida, India is a record of the candidate own work carried out by him/them under my supervision. The matter embodied in this thesis is original and has not been submitted for the award of any other degree

Signature of Examiner

Signature of Supervisor

Date: 11 June, 2026

Place: Greater Noida

ACKNOWLEDGEMENT

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Signature:

*Name : **ANKIT PATEL***

*Roll No.:**23131010430***

*Date :**11 JUNE 2026***

ABSTRACT

The rapid growth of internet technologies has increased the demand for efficient and real-time communication systems. Traditional messaging platforms often face limitations in terms of scalability, responsiveness, and real-time synchronization. To address these challenges, **SamvaadX: Real-Time Chat Application Using Node.js and Socket.IO** has been developed as a modern web-based communication platform that enables users to exchange messages instantly over the internet.

The primary objective of this project is to provide a secure, fast, and user-friendly chat application capable of supporting real-time messaging between multiple users. The system is developed using the MERN stack technologies, including React.js for the frontend, Node.js and Express.js for the backend, MongoDB for database management, and Socket.IO for real-time bidirectional communication. User authentication and authorization mechanisms are implemented to ensure secure access to the platform.

The application allows users to create accounts, log in securely, search for other users, view online/offline status, send and receive messages instantly, and log out safely. Socket.IO establishes a persistent connection between the client and server, enabling immediate message delivery without requiring page refreshes. MongoDB is used to store user information and chat data efficiently.

The implementation and testing of SamvaadX demonstrate that the system successfully achieves low-latency communication, reliable message delivery, and an intuitive user experience. The application performs efficiently under normal usage conditions and provides a scalable foundation for future enhancements.

Future developments may include group chat functionality, voice and video calling, file sharing, push notifications, end-to-end encryption, and AI-powered chat assistance. Overall, SamvaadX serves as an effective real-time communication solution that showcases the practical application of modern web technologies in developing interactive and scalable messaging platforms.

INTRODUCTION

Traditional communication systems often suffer from delays, limited real-time interaction, and inefficient message synchronization between users. There is a need for a fast, secure, and reliable platform that enables users to exchange messages instantly and monitor user availability in real time. The challenge is to develop a scalable chat application that provides seamless communication with low latency and an intuitive user experience. SamvaadX addresses these challenges by leveraging Node.js, Socket.IO, React.js, and MongoDB to deliver real-time messaging and user connectivity.

Problem Introduction:

Modern users require instant communication, but traditional web applications often experience message delays and lack real-time interaction. SamvaadX solves this problem by providing a secure and real-time chat platform using Node.js, Socket.IO, React.js, and MongoDB.

Motivation:

Growing dependence on online communication motivates the development of efficient real-time messaging applications.

Project Objectives:

1. Real-time communication.
2. Secure authentication.
3. User-friendly interface.
4. Presence detection.
5. Scalable architecture.

Scope:

SamvaadX provides a secure and real-time messaging platform for users. It supports instant communication, user authentication, and online status tracking. Future enhancements can include group chats, voice/video calls, and file sharing.

Related Work:

Applications such as WhatsApp, Telegram, Discord, and Messenger demonstrate the importance of low-latency communication.

LITERATURE SURVEY

2.1 WhatsApp

- Popular real-time messaging application.
- Provides instant text, voice, and video communication.
- Uses end-to-end encryption for security.
- Supports group chats and media sharing.

2.2 Telegram

- Cloud-based messaging platform.
- Fast message synchronization across devices.
- Supports large groups and channels.
- Provides high scalability and security features.

2.3 Discord

- Real-time communication platform for communities.
- Supports text, voice, and video chat.
- Provides low-latency communication.
- Widely used for collaborative discussions.

2.4 Socket.IO Technology

- Enables real-time bidirectional communication.
- Uses WebSocket protocol for instant message delivery.
- Reduces communication delay.
- Suitable for chat applications.

2.5 MERN Stack Technology

- MongoDB for database management.
- Express.js for server-side development.
- React.js for user interface design.
- Node.js for backend processing.

Summary

- Existing chat applications provide real-time communication and user-friendly interfaces.
- Socket.IO is an effective technology for instant messaging.
- MERN Stack offers scalability, flexibility, and high performance.
- SamvaadX combines these technologies to build a secure and efficient real-time chat application.

SOFTWARE REQUIREMENTS SPECIFICATION

3.1 Product Perspective

SamvaadX is a web-based real-time chat application based on a client-server architecture.

3.2 System Interfaces

Browser ↔ Node.js Server ↔ MongoDB Database.

3.3 User Interfaces

- Login Page
- Registration Page
- Chat Dashboard

3.4 Hardware Interfaces

No special hardware is required. A computer or smartphone with internet access is sufficient.

3.5 Software Interfaces

- Node.js
- Express.js
- Socket.IO
- MongoDB
- React.js

3.6 Communication Interfaces

WebSocket communication using Socket.IO over HTTP/HTTPS.

3.7 Product Functions

- User Registration
- User Login
- Real-Time Messaging
- Online Status Tracking
- User Search
- Logout

3.8 User Characteristics

Users should have basic knowledge of computers and internet usage.

3.9 Constraints

- Stable Internet Connection
- Browser Compatibility
- Database Availability

3.10 Assumptions

- Server remains available.
- Users have active internet connectivity.

3.11 Use Cases

- Register
- Login
- Send Message
- Receive Message
- Logout

SYSTEM DESIGN, IMPLEMENTATION AND RESULTS

Architecture:

Client Layer, Application Layer, Database Layer.

Database Design:

Users Collection and Messages Collection.

Implementation:

Frontend developed using HTML, CSS, and JavaScript.

Backend developed using Node.js and Express.js.

Real-time messaging implemented through Socket.IO.

Testing:

Functional testing performed on registration, login, messaging, and logout modules.

Results:

Successful instant message delivery with online/offline status updates.

Interface Snapshots

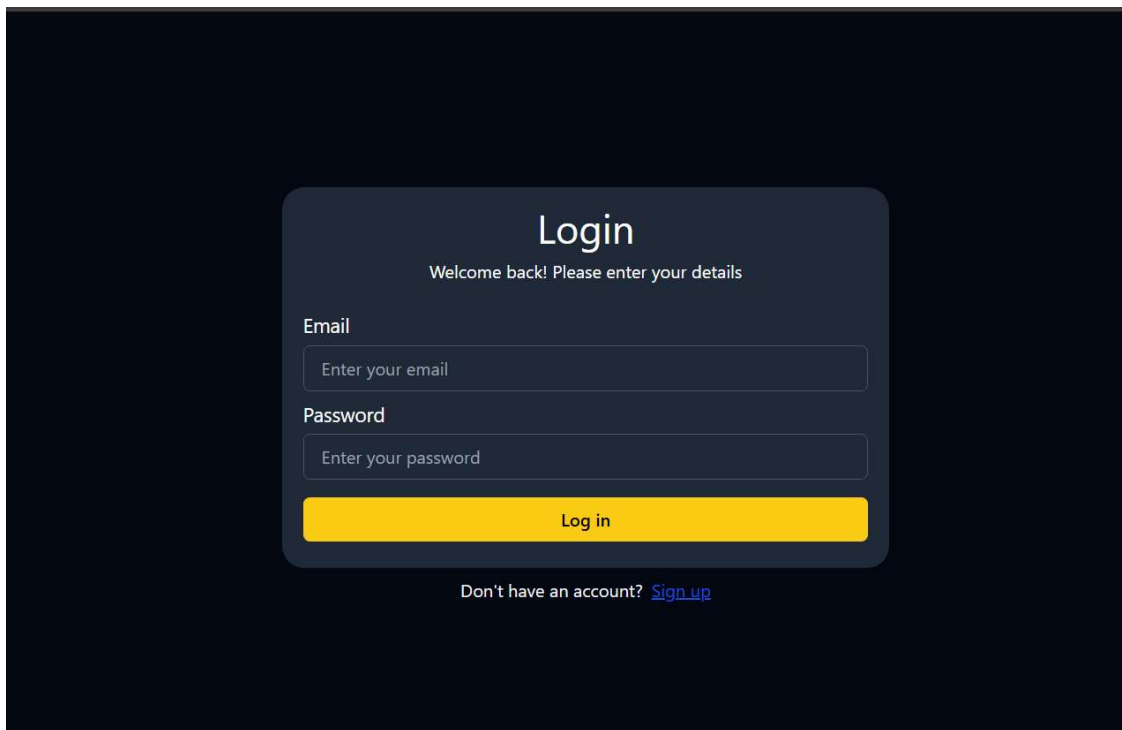
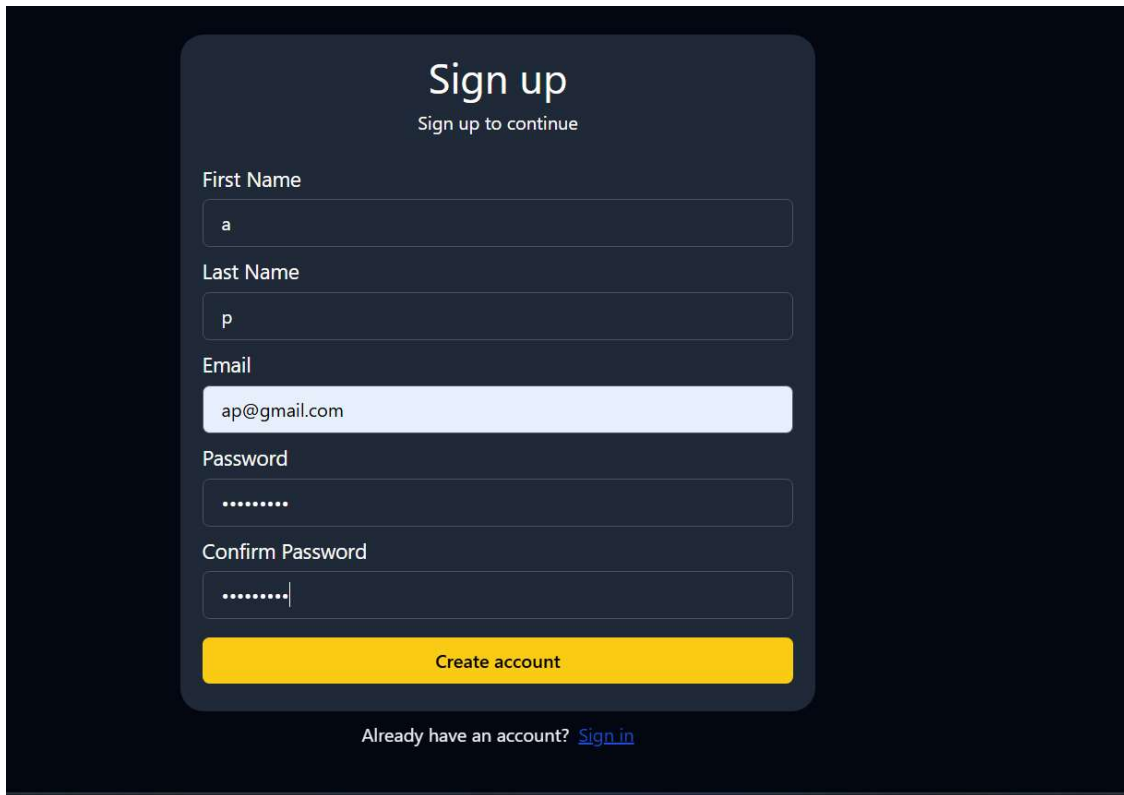


Figure 4.1 Login Page



The registration page features a dark blue background with a central white rounded rectangle containing the form. The form is titled 'Sign up' in a large, bold, black font, with the subtitle 'Sign up to continue' in a smaller, regular black font below it. The form includes five input fields: 'First Name' with the letter 'a', 'Last Name' with the letter 'p', 'Email' with 'ap@gmail.com', 'Password' with seven dots, and 'Confirm Password' with seven dots. A prominent red 'Create account' button is positioned below the password fields. At the bottom of the form, a link reads 'Already have an account? [Sign in](#)'.

Sign up

Sign up to continue

First Name

a

Last Name

p

Email

ap@gmail.com

Password

.....

Confirm Password

.....

Create account

Already have an account? [Sign in](#)

Figure 4.2 Registration Page

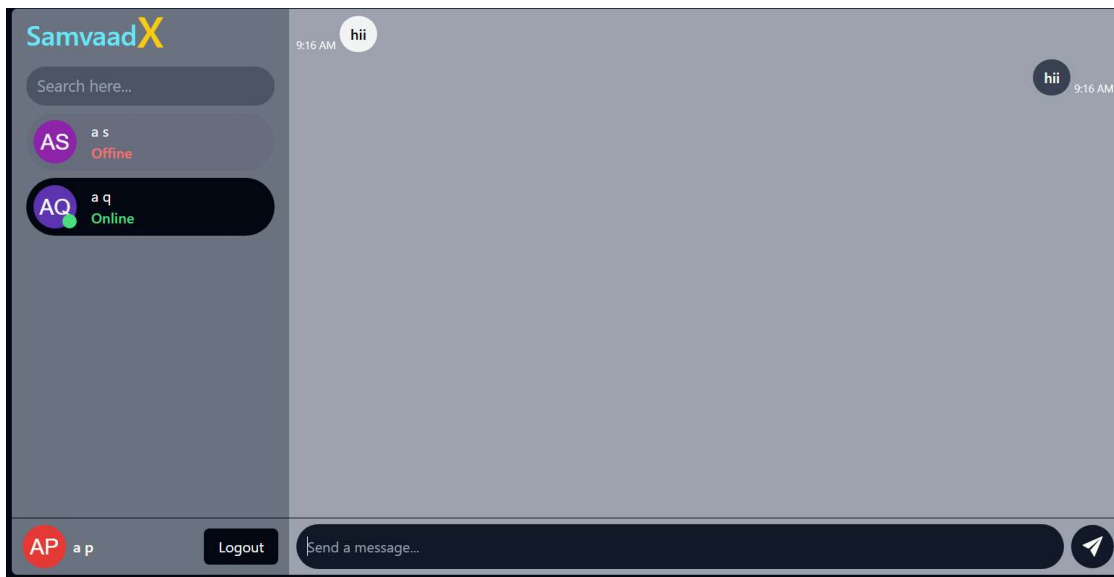


Figure 4.3 Chat Interface

CONCLUSION

SamvaadX successfully implements a secure and efficient real-time chat environment. The use of Socket.IO enables low-latency communication, while Node.js ensures scalability and performance. The project fulfills its objectives and demonstrates practical implementation of modern web development technologies.

Performance Evaluation

The application demonstrated reliable real-time communication with minimal delay during testing.

Future Scope

Future enhancements include:

- Group Chats
- File and Media Sharing
- End-to-End Encryption
- Push Notifications
- Mobile Application Development
- Voice and Video Calling

REFERENCES

Node.js Documentation – <https://nodejs.org>

Socket.IO Documentation – <https://socket.io>

MongoDB Documentation – <https://mongodb.com>

Express.js Documentation – <https://expressjs.com>